

Project Title:

Machine Learning Framework for Urban Traffic Accident Risk Prediction using Text Analysis

Abstract

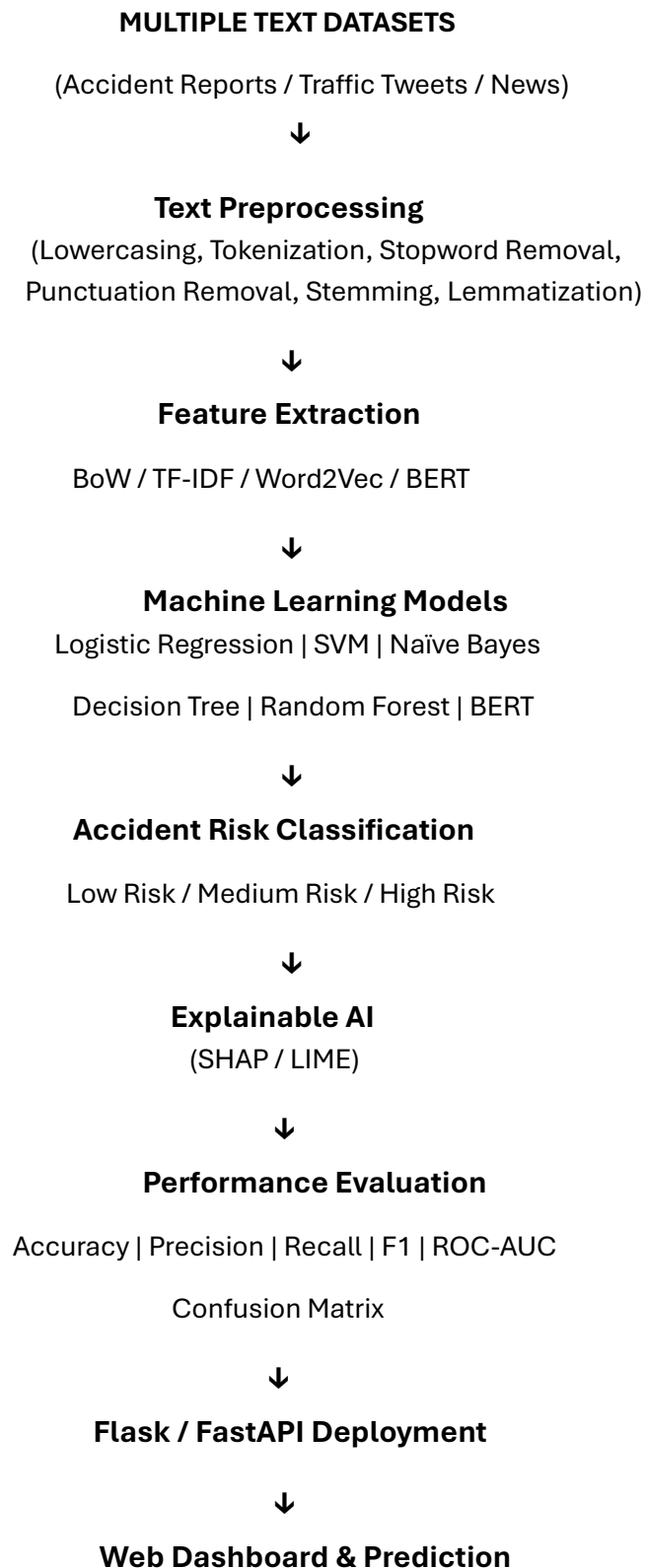
Urban traffic accidents are a significant public safety concern, resulting in loss of lives, injuries, traffic congestion, and economic damage. A large amount of accident-related information is generated daily through police reports, traffic incident descriptions, emergency service logs, news articles, and social media platforms. However, traditional traffic management systems mainly rely on structured data and provide limited support for extracting meaningful insights from unstructured textual information. This project proposes a **Machine Learning Framework for Urban Traffic Accident Risk Prediction using Text Analysis** that utilizes Natural Language Processing (NLP) and machine learning techniques to analyze traffic incident reports and predict accident risk levels.

The proposed system performs **text preprocessing**, including tokenization, lowercasing, stop-word removal, punctuation removal, stemming, and lemmatization. Text features are extracted using **Bag-of-Words (BoW)**, **TF-IDF**, and **Word Embeddings**, while contextual representations are generated using **BERT**. Multiple supervised machine learning algorithms such as **Logistic Regression**, **Support Vector Machine (SVM)**, **Naïve Bayes**, **Decision Tree**, **Random Forest**, and **BERT-based classifiers** are trained and evaluated to classify traffic incidents into different accident risk categories, including **Low Risk**, **Medium Risk**, and **High Risk**.

To improve model transparency, **Explainable AI (XAI)** techniques such as **SHAP** and **LIME** are incorporated to identify the important words and phrases that influence the model's predictions. The performance of the proposed framework is evaluated using **Accuracy**, **Precision**, **Recall**, **F1-score**, **ROC-AUC score**, and **Confusion Matrix**. The final model is deployed using **Flask/FastAPI**, providing a user-friendly web interface for accident risk prediction. The proposed framework aims to assist traffic management authorities in identifying accident-prone situations, improving road safety, supporting proactive decision-making, and contributing to intelligent transportation and smart city initiatives.

Dataset	Text Available	URL
Road Accident Dataset (Kaggle)	Contains accident-related categorical/text fields/Weather fields	https://www.kaggle.com/datasets/xavierberge/road-accident-dataset?utm_source=chatgpt.com
Traffic Accidents Dataset (Kaggle)	Text classification	https://www.kaggle.com/search?q=twitter+traffic+accidents&utm_source=chatgpt.com

Workflow :



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